

CSE 340 Assignment 3

Deadline: 20 August 2026 11:30 PM

Submission link: <https://forms.gle/6xDD2zUfxVieo2yJ6>

Solve all five of the following questions:

1. Convert -834.12×10^4 to IEE 32-bit single precision floating-point representation. (You don't have to write the last 0 bits of fraction part, writing up to necessary bit is sufficient)
2. The following two numbers are represented in IEEE 32-bit single precision floating point convention.
0xC0F147AE and 0x42A43D71
Multiply them. You need to show all steps.
3. Consider the following imaginary 22-bit floating point system. Where 14 bit is reserved for fraction part. Find the maximum and minimum possible value of this system.
4. In floating point number system, how many numbers of the exponents are reserved. What are they reserved for? Write all use case of these reserved exponents.
5. Multiply two integer values, 0101 and 0011 using optimized multiplier. List the values of current product and multiplicand in each step.